



Faculty of Forestry, University of British Columbia
2nd Doctoral Committee Meeting

The fine balance

Exploring the Interplay of Reproductive Strategies and Growth Dynamics in Trees

Xiaomao (Mao) Wang

Supervisor: Elizabeth Wolkovich

Table of content

- Brief background
- Research questions
- Individual chapters
- Other activities

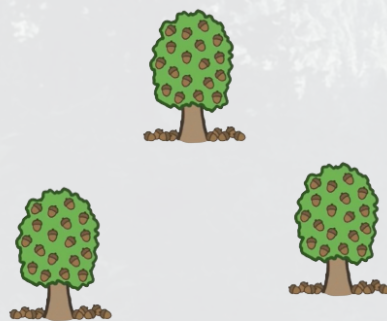


What is masting?

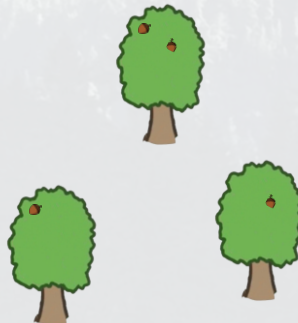
Synchronized variation of large seed crop within a population between years

Characteristics:

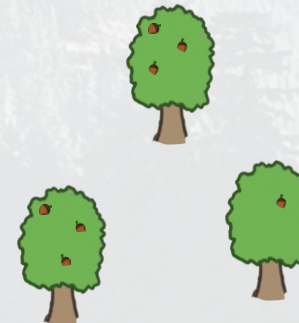
- Interannual variation
- Spatial synchrony
- Temporal autocorrelation
- Frequency



Mast year



Mast year + 1



Mast year + 2

Hypotheses explaining masting



Predator satiation



Pollination coupling



Resource matching



Environmental cues

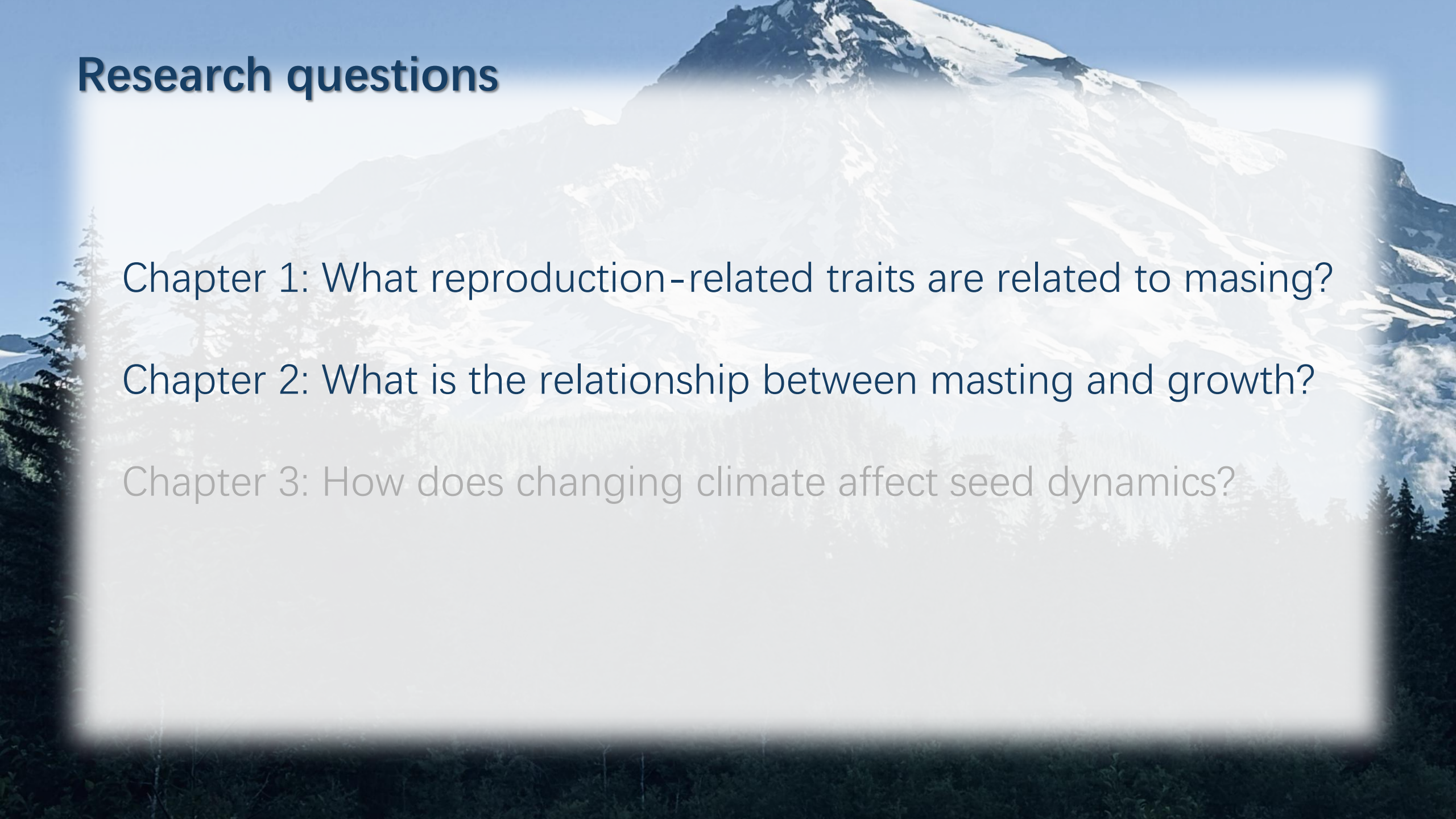
Research questions

Chapter 1: What reproduction-related traits are related to masting?

Chapter 2: What is the relationship between masting and growth?

Chapter 3: How does changing climate affect seed dynamics?

Research questions



Chapter 1: What reproduction-related traits are related to masting?

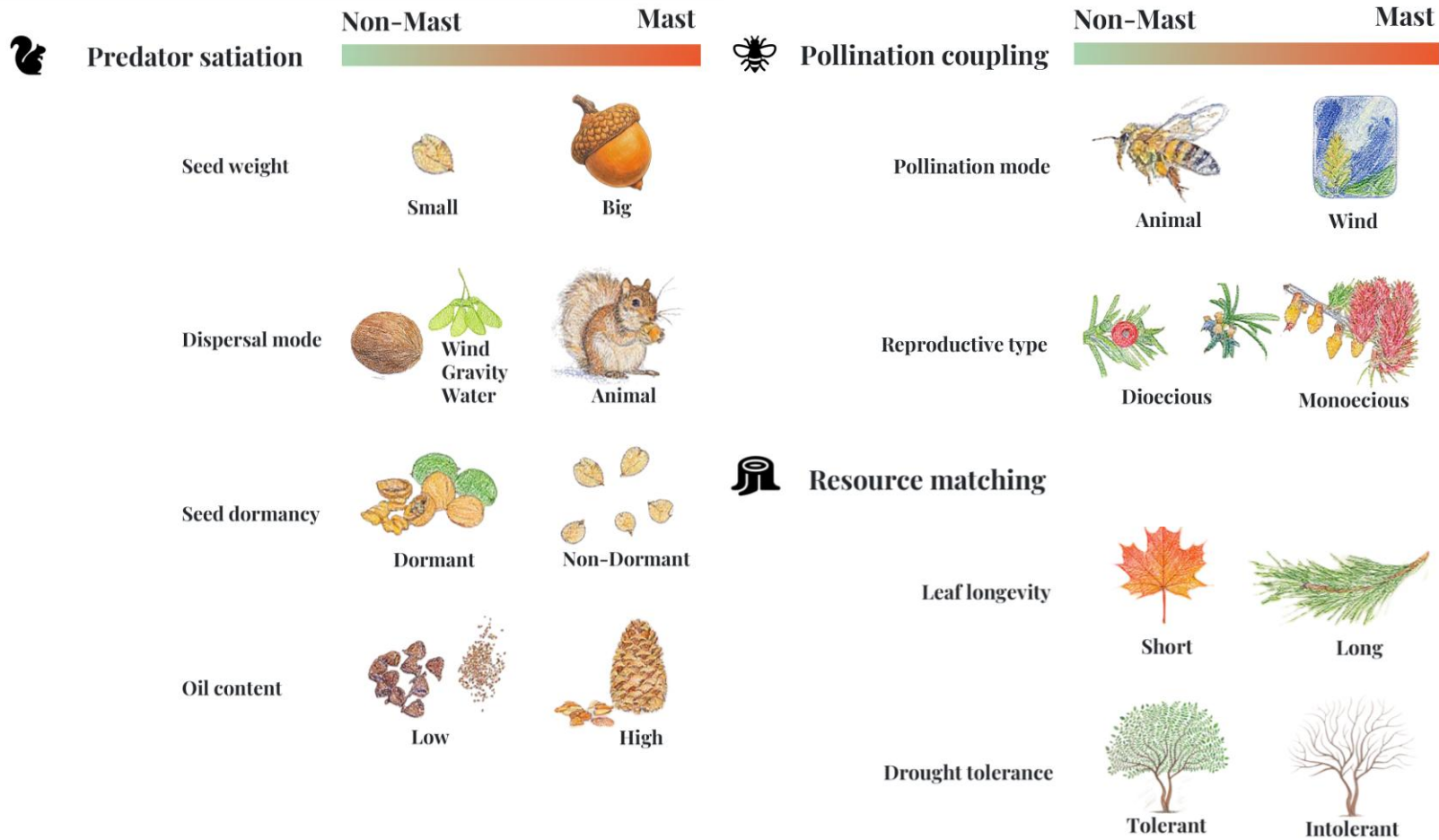
Chapter 2: What is the relationship between masting and growth?

Chapter 3: How does changing climate affect seed dynamics?

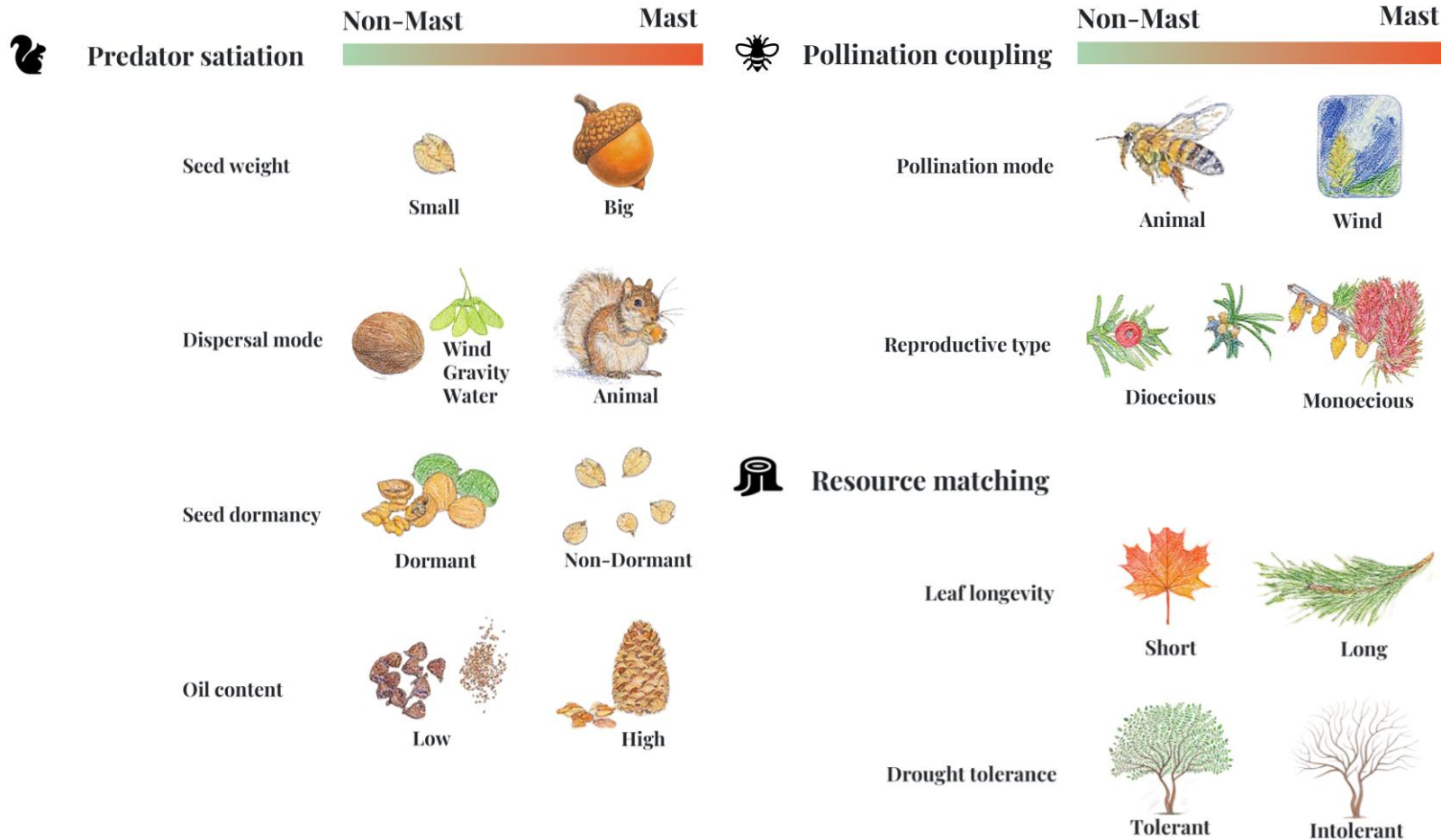
Chapter 1: Mast-Traits

- Masting can involve different, **non-mutually exclusive mechanisms** at different **reproductive stages**:
 - Pollination coupling (pollination)
 - Resource availability (seed maturation)
 - Predator satiation (seed dispersal)
- Species differ in **traits** that may influence these mechanisms.
- Trait–masting associations may therefore provide clues about the **evolutionary drivers** of masting.
- Trees in North America

Chapter 1: Mast-Traits



Chapter 1: Mast-Traits



Data:

Silvics of North America

Masting: strong or weak

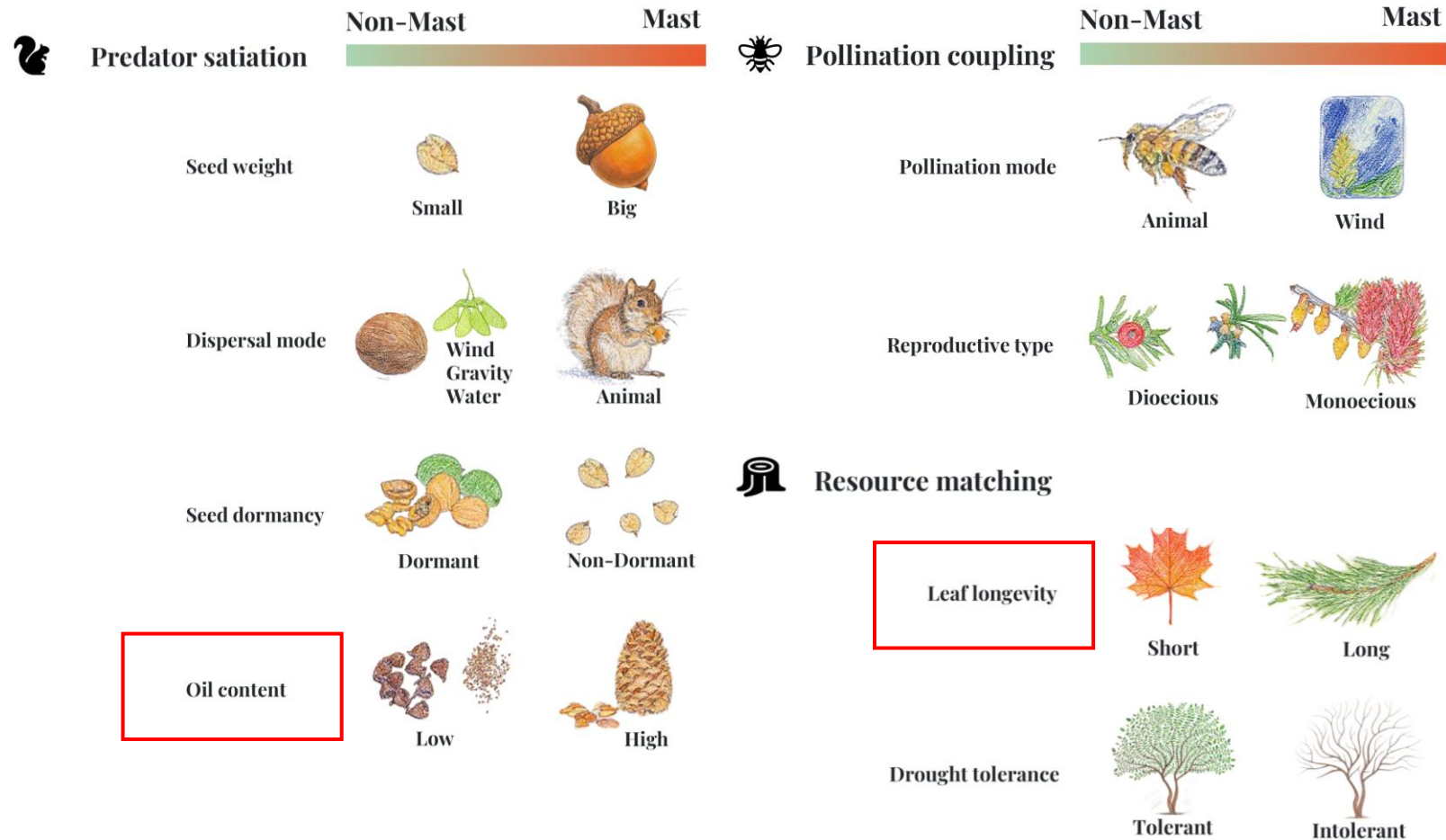
Angiosperm:

- 98 with masting data
- 51 strong masting

Gymnosperm:

- 61 with masting data
- 54 strong masting

Chapter 1: Mast-Traits



Data:

Silvics of North America

Masting: strong or weak

Angiosperm:

- 98 with masting data
- 51 strong masting

Gymnosperm:

- 61 with masting data
- 54 strong masting

Chapter 1: Mast-Traits

Current progress:

- Statistical analyses
 - Single trait with phyloglm
 - Multivariate analysis with NMDS
- Outlines of the paper

Completed

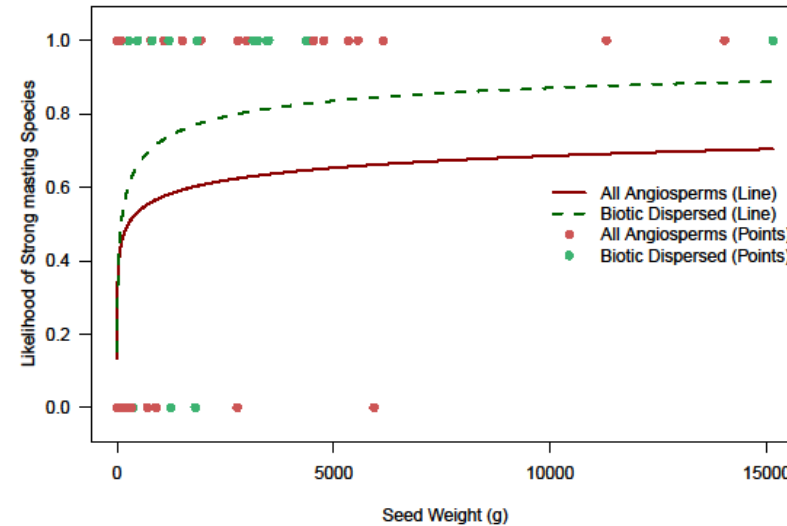
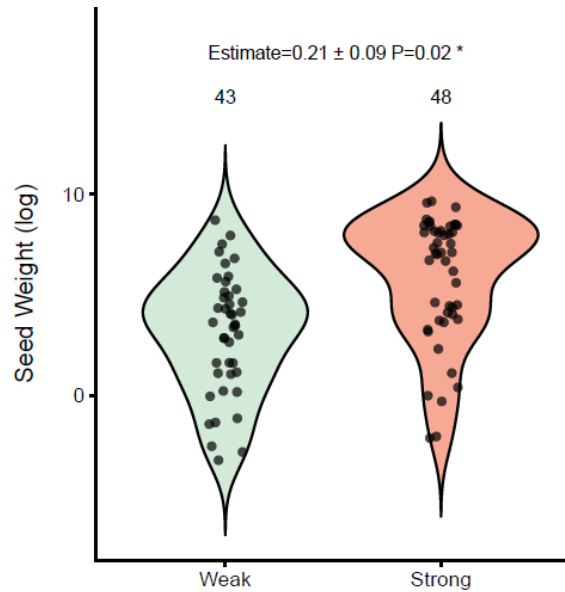
- Manuscript write-up

On going

- Finalize some discussion on evolutionary aspects

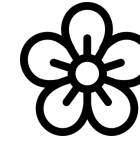
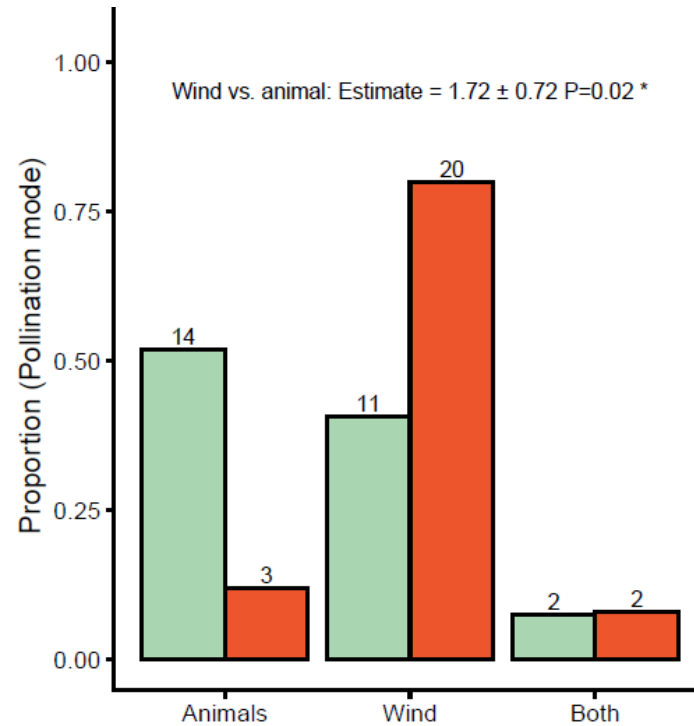
Need discussion

Chapter 1: Mast-Traits

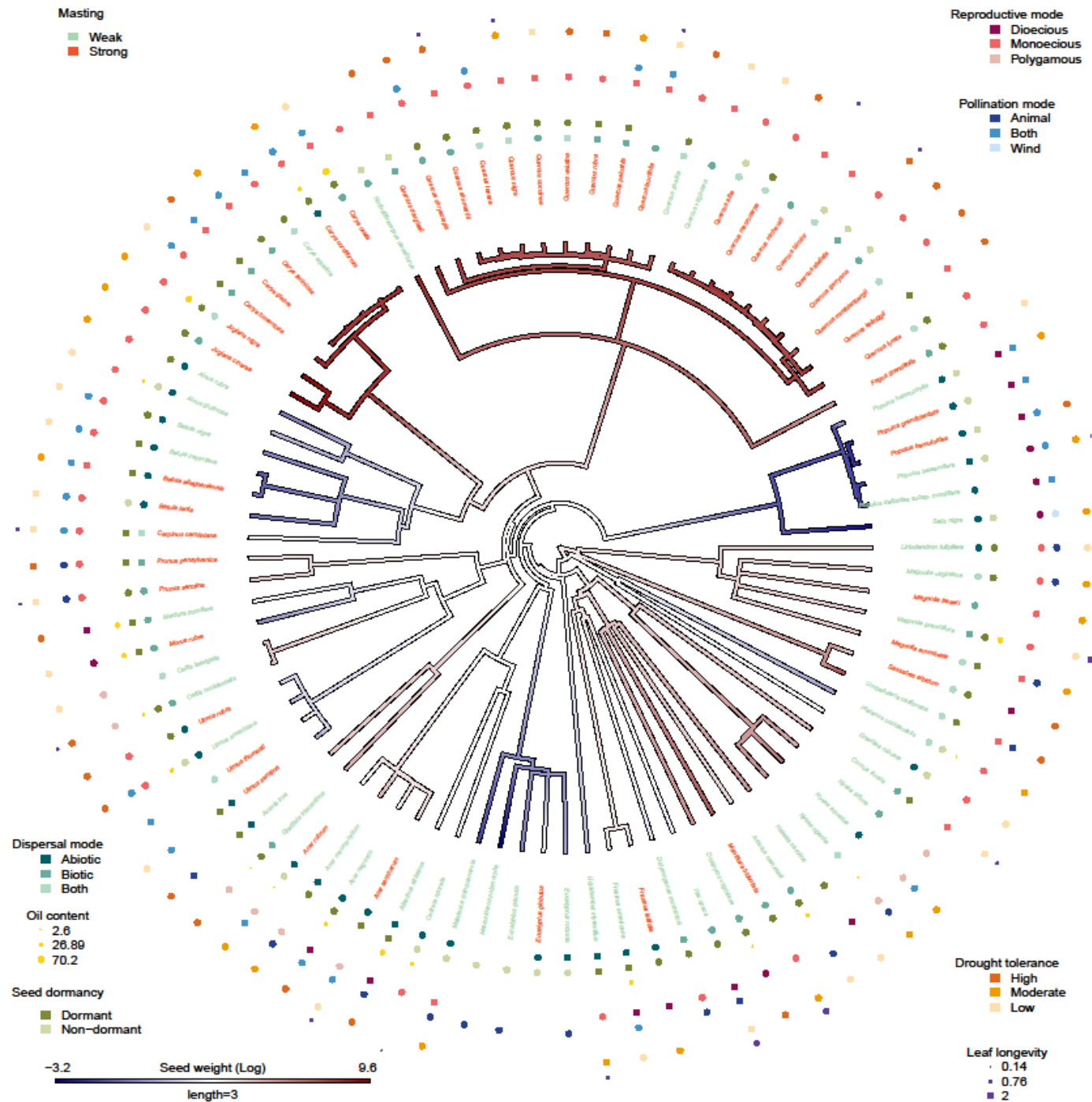


- Heavier seeds are associated with strong masting
- The effect was stronger among species with biotically dispersed seeds

Chapter 1: Mast-Traits



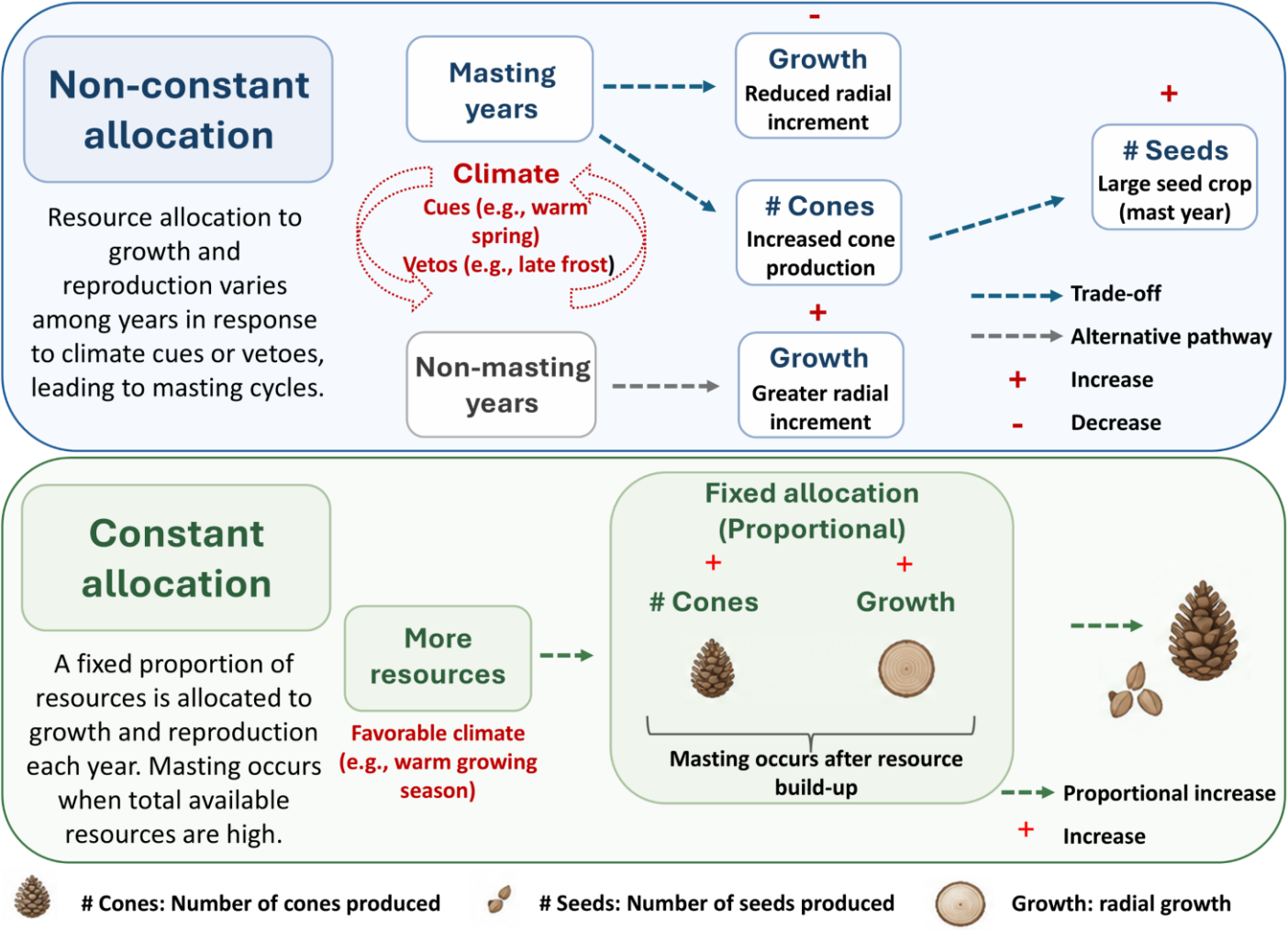
- Wind pollinated species are more likely to be strong masting
- All gymnosperms are wind-pollinated



Chapter 1: Mast-Traits

- Reproductive traits point to different stages of selection
 - Wind pollination → pollination coupling
 - Seed weight → predator satiation
 - Coordinated strategies across whole reproductive process
- Masting may be shaped differently across evolutionary contexts
 - Masting is population-level phenomenon, while traits were species-level
 - Strong phylogenetic structure of traits
 - Angiosperms vs. gymnosperms may represent different evolutionary pathways

Chapter 2: What's the relationship between growth and reproduction?



By Mao

Chapter 2: Growth-Reproduction Trade-off

Current progress:

- Core samples scanning
- Cores measurement for 6 stands

Completed

- Cross-dating for cores
- Building the model

On going



Chapter 2: Growth-Reproduction Trade-off

Growth

$$\mathbf{BAI}_{i,t} \sim \text{Normal}(\alpha_{\text{BAI}} + \beta_{1,\text{SM}} * \text{Soil moisture}_t, \sigma_{\text{BAI}})$$

$$G_{i,t} = \mathbf{BAI}_{i,t} - \alpha_{\text{BAI}} - \beta_{1,\text{SM}} * \text{Soil moisture}_t$$

$$G_t = \text{mean}(G_{i,t})$$

Chapter 2: Growth-Reproduction Trade-off

Growth

$$\text{BAI}_{i,t} \sim \text{Normal}(\alpha_{\text{BAI}} + \beta_{1,\text{SM}} * \text{Soil moisture}_t, \sigma_{\text{BAI}})$$

$$\mathbf{G}_{i,t} = \text{BAI}_{i,t} - \alpha_{\text{BAI}} - \beta_{1,\text{SM}} * \text{Soil moisture}_t$$

$$\mathbf{G}_t = \text{mean}(\mathbf{G}_{i,t})$$

Chapter 2: Growth-Reproduction Trade-off

Growth

$$\text{BAI}_{i,t} \sim \text{Normal}(\alpha_{\text{BAI}} + \beta_{1,\text{SM}} * \text{Soil moisture}_t, \sigma_{\text{BAI}})$$

$$G_{i,t} = \text{BAI}_{i,t} - \alpha_{\text{BAI}} - \beta_{1,\text{SM}} * \text{Soil moisture}_t$$

$$\mathbf{G}_t = \text{mean}(G_{i,t})$$

Chapter 2: Growth-Reproduction Trade-off

Growth

$$\text{BAI}_{i,t} \sim \text{Normal}(\alpha_{\text{BAI}} + \beta_{1,\text{SM}} * \text{Soil moisture}_t, \sigma_{\text{BAI}})$$

$$G_{i,t} = \text{BAI}_{i,t} - \alpha_{\text{BAI}} - \beta_{1,\text{SM}} * \text{Soil moisture}_t$$

$$G_t = \text{mean}(G_{i,t})$$

Reproduction

$$\mu_{\text{sc},t} = \alpha_{\text{sc}} + \gamma_{\text{current}} \cdot G_t + \gamma_{\text{lag}} \cdot G_{t-1}$$

$$\text{sc}_t \sim \text{LogNormal}(\mu_{\text{sc},t}, \sigma_{\text{sc}})$$

Chapter 2: Growth-Reproduction Trade-off

Growth

$$\text{BAI}_{i,t} \sim \text{Normal}(\alpha_{\text{BAI}} + \beta_{1,\text{SM}} * \text{Soil moisture}_t, \sigma_{\text{BAI}})$$

$$G_{i,t} = \text{BAI}_{i,t} - \alpha_{\text{BAI}} - \beta_{1,\text{SM}} * \text{Soil moisture}_t$$

$$G_t = \text{mean}(G_{i,t})$$

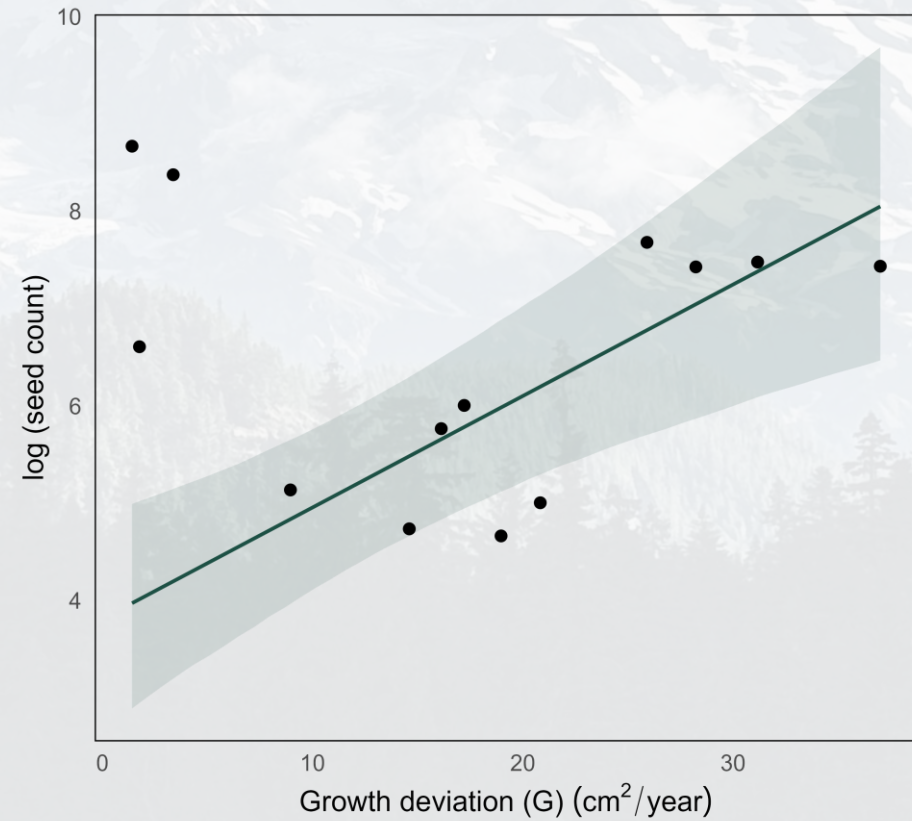
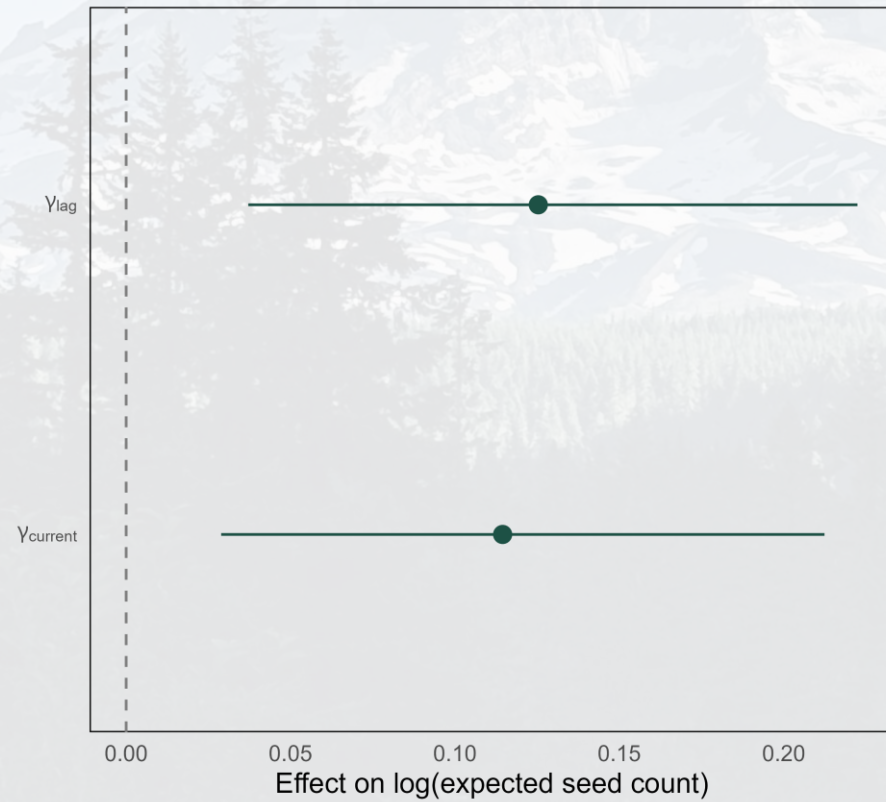
Reproduction

$$\mu_{\text{sc},t} = \alpha_{\text{sc}} + \gamma_{\text{current}} \cdot G_t + \gamma_{\text{lag}} \cdot G_{t-1}$$

$$\text{sc}_t \sim \text{LogNormal}(\mu_{\text{sc},t}, \sigma_{\text{sc}})$$

Chapter 2: Growth-Reproduction Trade-off

One species (*Tsuga heterophylla*) at one stand (21 trees), from 2013 to 2024 (11 years):



Other activities

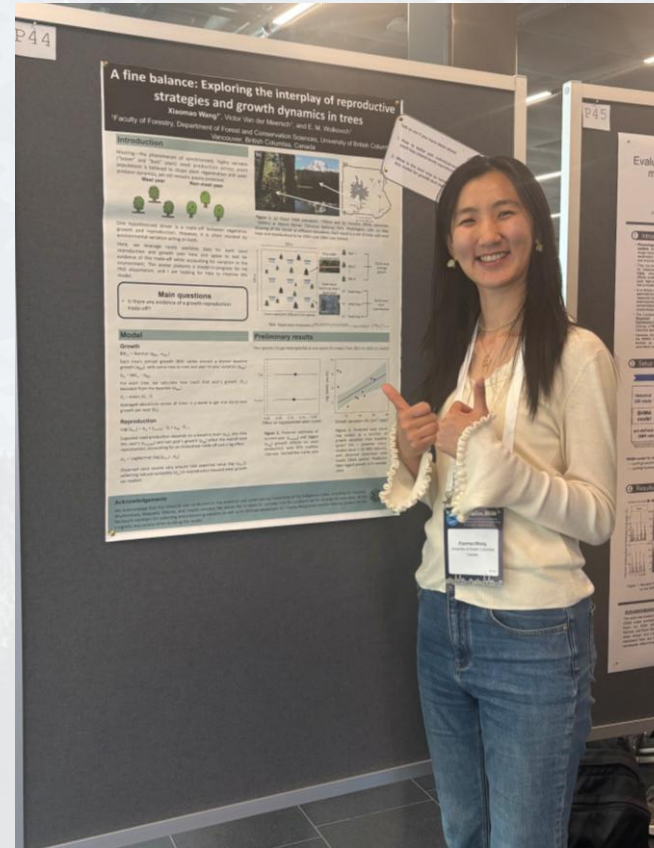


Fieldwork at Smithers



Fieldwork at Mount Rainier

Other activities



Poster presentation at StanCon 2026, Uppsala, Sweden

